



National Technical University of Athens
School of Electrical and Computer Engineering (ECE)

9th semester

Electric Machine Design

*Semester Project: Design of a Six-Pole Permanent-Magnet
Synchronous Motor*

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A Analytical Approach

A.1 Graphical Representation of the Winding

In Figure 1, the graphical representation of the winding configuration specified for the machine is presented. Specifically, the machine has 36 slots and a double-layer lap winding with a fractional pitch of $5/6$.

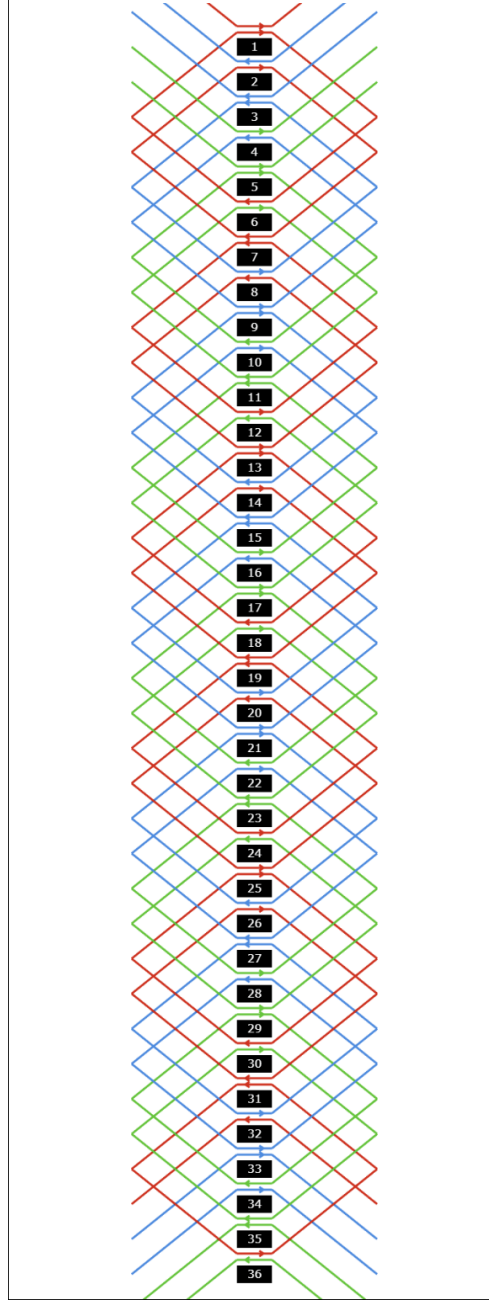


Figure 1: Winding configuration

A.2 Air-Gap MMF Distribution

The magnetomotive force (MMF) of one winding layer within the slot is calculated from:

$$F_{layer} = \sqrt{2} \cdot J_{N,rms} \cdot A_{layer} \cdot ff \cdot \cos(\varphi) \quad (1)$$

where:

- $J_{N,rms} = 4 \text{ A/mm}^2$: The RMS value of the nominal current density.
- $A_{layer} = 18.037 \text{ mm}^2$: The area of one slot layer, as calculated in FEMM.
- $ff = 0.7$: The fill factor.
- φ : The current phase angle.

For the instant at which the current of phase A reaches its maximum value, the phase angles φ for the three phases are defined as follows:

- Phase A: $\varphi = 0^\circ$
- Phase B: $\varphi = -120^\circ$
- Phase C: $\varphi = 120^\circ$

The phase distribution in the slots for one pole pair, for a double-layer winding with a pitch of 5/6, is presented in the following table:

Slot	0	1	2	3	4	5	6	7	8	9	10	11
Upper layer	A	A	-C	-C	B	B	-A	-A	C	C	-B	-B
Lower layer	A	-C	-C	B	B	-A	-A	C	C	-B	-B	A

Table 1: Phase distribution in the two layers per slot

The total air-gap MMF is obtained by summing the contributions of the slot currents. The resulting waveform has a stepped shape, which is repeated for the three pole pairs of the machine, as shown in Figure 2.

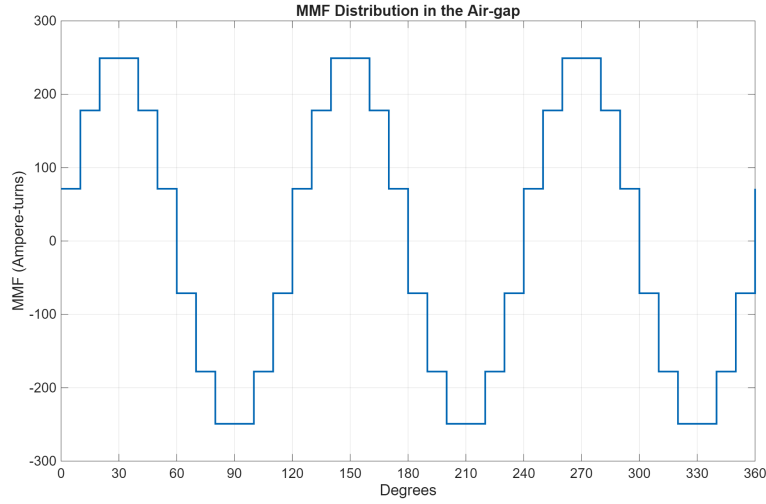


Figure 2: MMF distribution

A.3 Calculation of Winding Factors

Distribution Factor (k_b)

The number of slots per pole per phase for the considered machine is:

$$q = \frac{36}{6 \cdot 3} = 2 \quad (2)$$

The slot pitch in electrical degrees is:

$$\gamma = \frac{180^\circ \cdot 6}{36} = 30^\circ \quad (3)$$

The winding distribution factor is given by:

$$k_{b,n} = \frac{\sin\left(\frac{qn\gamma}{2}\right)}{q \sin\left(\frac{n\gamma}{2}\right)} \quad (4)$$

For the fundamental harmonic of the machine:

$$k_{b,1} = \frac{\sin\left(\frac{2 \cdot 30^\circ}{2}\right)}{2 \sin\left(\frac{30^\circ}{2}\right)} = \frac{\sin(30^\circ)}{2 \sin(15^\circ)} = 0.9659 \quad (5)$$

Pitch Factor (k_p)

The fractional pitch of 5/6 corresponds to an electrical angle of:

$$\rho = \frac{5}{6} \cdot 180^\circ = 150^\circ \quad (6)$$

The winding pitch factor is given by:

$$k_{p,n} = \cos\left(\frac{n(180^\circ - \rho)}{2}\right) \quad (7)$$

For the fundamental harmonic of the machine:

$$k_{p,1} = \cos\left(\frac{180^\circ - 150^\circ}{2}\right) = \cos(15^\circ) = 0.9659 \quad (8)$$

Winding Factor (k_w)

The overall winding factor is obtained as the product of the two factors above:

$$k_{w,n} = k_{b,n} \cdot k_{p,n} = \frac{\sin\left(\frac{qn\gamma}{2}\right)}{q \sin\left(\frac{n\gamma}{2}\right)} \cdot \cos\left(\frac{n(180^\circ - \rho)}{2}\right) \quad (9)$$

For the fundamental harmonic of the machine:

$$k_{w,1} = k_{b,1} \cdot k_{p,1} = 0.9659 \cdot 0.9659 = 0.933 \quad (10)$$

A.4 Calculation of the Fundamental and Harmonic Field Components

Assuming a smooth rotor, the magnetic flux density B_n is calculated from:

$$B_n = \mu_0 \cdot \frac{F_n}{l_{g,eff}} \quad (11)$$

where:

- $\mu_0 = 4\pi \cdot 10^{-7}$ H/m.
- F_n : The amplitude of the n -th harmonic of the MMF.
- $l_{g,eff}$: The effective air-gap length.

MMF Calculation

The amplitude of the n -th harmonic component of the air-gap magnetomotive force (MMF), per pole pair, for a three-phase winding, is given by:

$$F_n = \frac{1}{n} \cdot \frac{3}{2} \cdot \frac{4}{\pi} \cdot k_{w,n} \cdot \frac{N_\phi}{p} \cdot \sqrt{2} \cdot I_N \quad (12)$$

where:

- $k_{w,n}$: The winding factor for the n -th harmonic, calculated from Equation (9).
- $N_\phi = 2N = 30$: The number of conductors per phase and per slot.
- $p = 3$: The number of pole pairs of the machine.
- I_N : The nominal RMS current, calculated from the net copper cross-sectional area per layer:

$$I_N = J_{N,rms} \cdot \frac{A_{layer} \cdot ff}{N} = 4 \cdot \frac{18.037 \cdot 0.7}{15} \approx 3.367 \text{ A} \quad (13)$$

Calculation of the Effective Air-Gap Length

For an accurate calculation of the air-gap flux density, the non-uniformity introduced by the stator slots must be taken into account. To calculate the field, the effective air-gap length ($l_{g,eff}$) is used, which is obtained from the physical air-gap length $l_g = 0.69 \text{ mm}$ using the Carter coefficient (k_C):

$$l_{g,eff} = k_C \cdot l_g \quad (14)$$

The Carter coefficient is calculated from:

$$k_C = \frac{\tau_u}{\tau_u - b_e} = \frac{\tau_u}{\tau_u - \kappa b_1} \quad (15)$$

where the symbols used are defined as follows:

- b_e : The equivalent slot-opening width over which the flux density is assumed to be zero.
- $\tau_u = \frac{2 \cdot (33 + 3.9 + 0.69)}{36} = 6.561 \text{ mm}$: The slot pitch.
- $b_1 = 1.575 \text{ mm}$: The slot-opening width.
- κ : A coefficient calculated from:

$$\kappa = \frac{2}{\pi} \left(\arctan \frac{b_1}{2l_g} - \frac{2l_g}{b_1} \ln \sqrt{1 + \left(\frac{b_1}{2l_g} \right)^2} \right) \approx \frac{\frac{b_1}{l_g}}{5 + \frac{b_1}{l_g}} = \frac{\frac{1.575}{0.69}}{5 + \frac{1.575}{0.69}} = 0.313 \quad (16)$$

Using Equations (14) and (15), the effective air-gap length is calculated as:

$$l_{g,eff} = \frac{6.561}{6.561 - 0.313 \cdot 1.575} \cdot 0.69 = 0.746 \text{ mm} \quad (17)$$

Field Calculation

Combining Equations (11), (12), and (9), the amplitude of the fundamental field harmonic is calculated, and the amplitudes of the first 10 harmonics are presented in the following table.

The following observations can be made regarding the results in Table 2:

- **Even harmonics:** Even harmonics ($n = 2, 4, 6, \dots$) were not included in the calculations because the MMF distribution exhibits half-wave symmetry. In such cases, the Fourier series contains only odd harmonics.
- **Triplen harmonics:** Although the values of the triplen harmonics ($n = 3, 9, 15, \dots$) were calculated in the table, in a three-phase machine these harmonics cancel each other. Therefore, they do not contribute to the formation of a rotating air-gap field and do not produce torque.

n	$k_{w,n}$	F_n (At)	B_n (T)
1	0.9330	84.849	0.14290
3	0.5000	15.157	0.02553
5	0.0670	1.218	0.00205
7	0.0670	0.870	0.00147
9	0.5000	5.052	0.00851
11	0.9330	7.714	0.01299
13	0.9330	6.527	0.01099
15	0.5000	3.031	0.00511
17	0.0670	0.358	0.00060
19	0.0670	0.321	0.00054

Table 2: Harmonic Content of Air-gap MMF and Magnetic Flux Density

A.5 Calculation of Leakage Inductance

To calculate the leakage inductance, the end windings and the effect of the permanent magnets are neglected, and the focus is placed on slot leakage. The total inductance is obtained by summing the individual inductances of four distinct regions of the slot geometry, as shown in Figure 3. Specifically, the slot was divided into two distinct regions due to the fill factor: one representing copper and the other assumed to be air.

For the following calculations, it is assumed that the slot is slightly different from the geometry created in FEMM: it has a rectangular shape, i.e., without fillets at the ends and without a gradual reduction in thickness as it approaches the air gap.

Based on Figure 3, the following dimensions are used:

- $h_{slot} = 8mm$: Total slot length.
- $h_1 = ff \cdot h_{slot} = 0.7 \cdot 8 = 5.6mm$: Height of the copper region.
- $h_2 = (1 - ff) \cdot h_{slot} = (1 - 0.7) \cdot 8 = 2.4mm$: Height of the air region.
- $h_3 = 1$: Height of the trapezoidal slot section.
- $h_4 = 1$: Height of the slot opening.
- $b = 1.575mm$: Slot-opening width.
- $b_s \approx \frac{5.36+3.89}{2} = 4.625$: Approximation of the theoretical slot width.

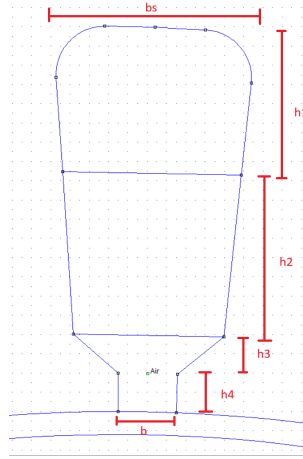


Figure 3: MMF distribution

Using the active machine length $L_s = 130$ mm and the number of conductors per slot $N_s = 2 \cdot N = 2 \cdot 15 = 30$, the inductances are calculated as follows:

- **Region 1 (Copper - L_1):** This region corresponds to the part of the slot occupied by the copper conductors:

$$L_1 = \mu_0 \cdot \frac{L_s}{b_s} \cdot \int_0^{h_1} \left(N_s \cdot \frac{x}{h_1}\right)^2 dx = N_s^2 \cdot L_s \cdot \mu_0 \cdot \frac{h_1}{3 \cdot b_s} = 30^2 \cdot 0.13 \cdot 4\pi \cdot 10^{-7} \cdot \frac{5.6}{3 \cdot 4.625} = 59.340 \mu H \quad (18)$$

- **Region 2 (Air - L_2):** This region corresponds to the part of the slot occupied by air:

$$L_2 = N_s^2 \cdot L_s \cdot \mu_0 \cdot \frac{h_2}{b_s} = 30^2 \cdot 0.13 \cdot 4\pi \cdot 10^{-7} \cdot \frac{2.4}{4.625} = 76.295 \mu H \quad (19)$$

- **Region 3 (Transition - L_3):** This is the trapezoidal section where the width varies from b_s to b :

$$L_3 = N_s^2 \cdot L_s \cdot \mu_0 \cdot \frac{h_3}{\frac{b_s + b}{2}} = 30^2 \cdot 0.13 \cdot 4\pi \cdot 10^{-7} \cdot \frac{1}{\frac{(4.625 + 1.575)}{2}} = 47.428 \mu H \quad (20)$$

- **Region 4 (Slot Opening - L_4):** This is the slot opening toward the air gap, with the minimum width b [?]:

$$L_4 = N_s^2 \cdot L_s \cdot \mu_0 \cdot \frac{h_4}{b} = 30^2 \cdot 0.13 \cdot 4\pi \cdot 10^{-7} \cdot \frac{1}{1.575} = 93.350 \mu H \quad (21)$$

The total leakage inductance of one slot is the sum of the above terms:

$$L_{slot, total} = L_1 + L_2 + L_3 + L_4 = 276.413 \mu H \quad (22)$$

Since the machine has 36 slots, the total leakage inductance per phase is:

$$L_{\sigma, ph} = \frac{36}{3} \cdot 217.073 \mu H = 3.317 mH \quad (23)$$

B Numerical Approach

PMSM Geometry in FEMM

The geometry of the permanent-magnet synchronous machine (PMSM) implemented in FEMM is presented below.

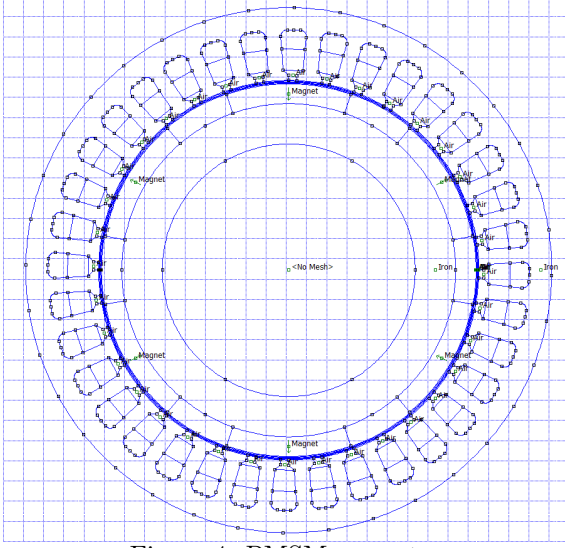


Figure 4: PMSM geometry

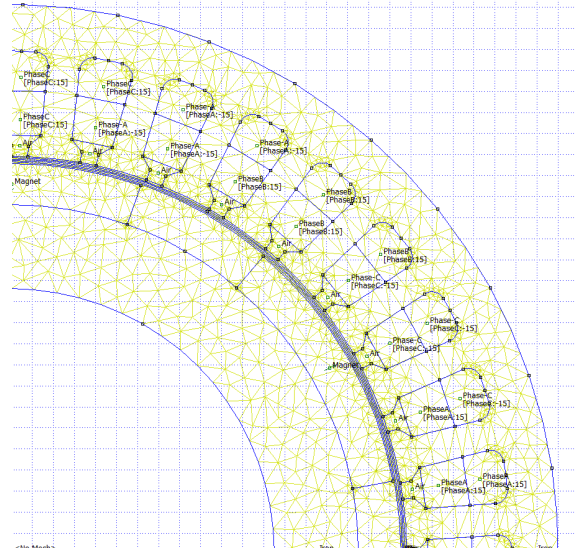


Figure 5: PMSM with mesh and circuits

B.1 D-axis Identification and T- δ Curve

Identification of the d-axis

In permanent-magnet synchronous machines (PMSMs), the analysis of operation is facilitated by the Clarke and Park transformations, which transfer the quantities from the stationary three-phase (abc) reference frame to a rotating two-axis ($d-q$) reference frame that rotates with the rotor.

The d -axis is defined as the axis coinciding with the direction of the magnetic flux produced by the rotor permanent magnets. The magnet flux (Ψ_m) is assumed to lie entirely along this axis. Specifically, the d -axis coincides with the position at which the phase-A flux linkage (Ψ_a) reaches its maximum. At this position, the magnetic axis of the rotor is fully aligned with the geometric axis of the phase-A coil.

In Figure 6, the phase-A flux linkage is shown over one complete electrical rotation of the rotor (i.e., 1/3 of a mechanical revolution). The flux reaches its maximum near the angle $offset = 69^\circ$, so this value is used as the offset.

Next, the rotor must be placed at an equilibrium position before the rotations begin. This position corresponds to the natural alignment of the magnets with the stator teeth, where the cogging torque is zero. This ensures that the torque ripple starts from a known reference, avoiding unwanted DC offsets in the final results. To determine this position, zero current was applied to the stator and the rotor was rotated with a very small step over the tooth corresponding to the maximum flux (i.e., from 65 to 75 degrees). Figure 7 was then generated, showing that the zero-torque position is $cogg = 68.75^\circ$.

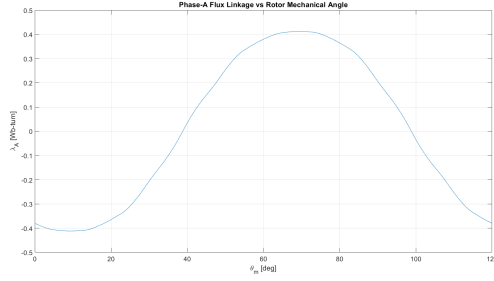


Figure 6: Flux A linkage

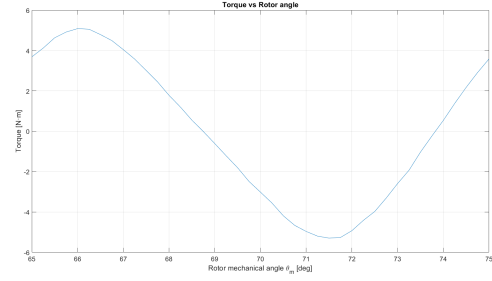


Figure 7: Cogging Torque

T- δ Curve

To calculate the electromagnetic torque (T_e) as a function of the internal torque angle (δ), two approaches were used.

The first approach, shown in Figure 8, consists of one complete electrical rotation of the stator currents at the nominal current density, i.e., $J_{N,rms} = 4A/mm^2$. The second approach, shown in Figure 9, consists of one complete electrical rotation of the rotor while keeping the nominal stator current constant.

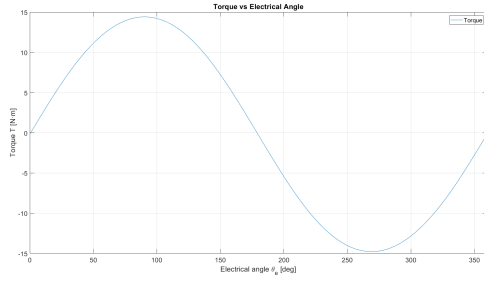


Figure 8: T- δ by rotating stator currents

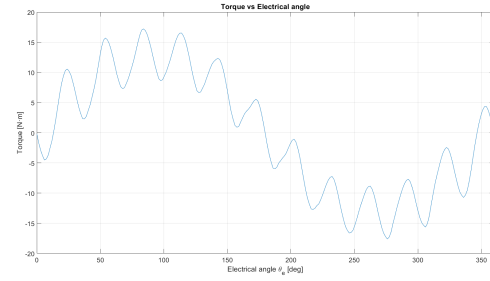


Figure 9: T- δ by rotating the rotor

The torque curve in Figure 9 deviates from the ideal sinusoidal shape. This distortion is caused by the continuous variation of the relative position between the rotor magnets and stator teeth during rotation. This variation changes the air-gap permeance, resulting in cogging torque that is superimposed on the total electromagnetic torque.

B.2 Magnetic Field Distribution for Different δ Angles

The following plots show the magnetic flux density B for different torque angles δ . The following operating conditions are considered:

- In Figure 10: no-load operation.
- In Figure 11: nominal current with an internal torque angle of 0° .
- In Figure 12: nominal current with an internal torque angle of 90° .
- In Figure 13: nominal current with an internal torque angle of 180° .

For the maximum cogging torque under no-load operation, as shown in Figure 7, $T_{cog,max} = 5.1Nm$.

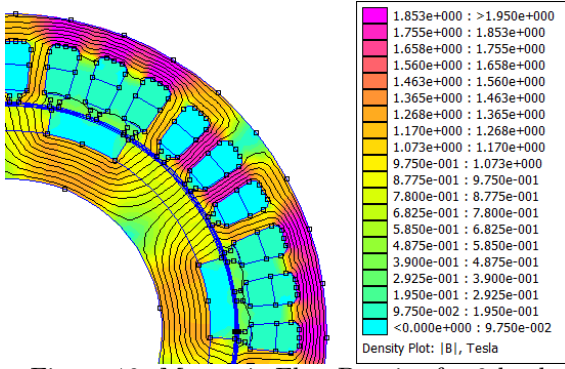


Figure 10: Magnetic Flux Density for 0 load

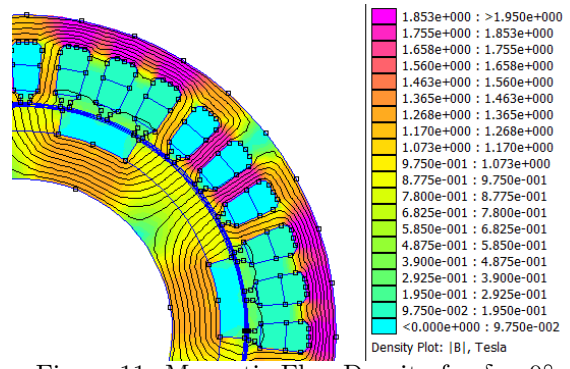


Figure 11: Magnetic Flux Density for $\delta = 0^\circ$

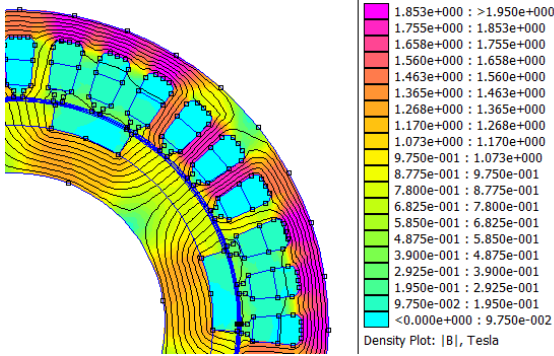


Figure 12: Magnetic Flux Density for $\delta = 90^\circ$

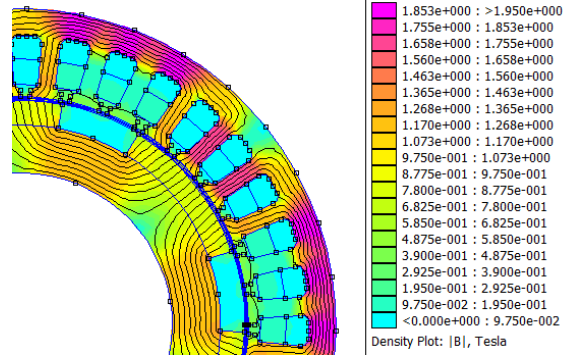


Figure 13: Magnetic Flux Density for $\delta = 180^\circ$

B.3 Calculation of Electromagnetic Torque, Flux Linkage, and Induced Voltage

The results obtained from the FEMM simulation for torque, RMS flux linkage, and induced voltage for all operating conditions of Question B.2 are presented in Table 3.

To determine the induced voltage in the machine windings, the rotor was rotated at constant synchronous speed. For a frequency of $f = 50$ Hz and $p = 3$ pole pairs, the synchronous mechanical speed n_s is defined as:

$$n_s = \frac{60 \cdot f}{p} = \frac{60 \cdot 50}{3} = 1000 \text{ rpm} \quad (24)$$

According to Faraday's law, the induced voltage in each phase is proportional to the rate of change of the flux linkage Ψ with respect to time t :

$$e(t) = -\frac{d\Psi(t)}{dt} = -\frac{d\Psi}{d\theta_e} \frac{d\theta_e}{dt} = -\omega_e \frac{d\Psi}{d\theta_e} \quad (25)$$

In this case, the flux linkage was assumed to have a sinusoidal waveform for simplicity.

Operating condition	T_e [Nm]	Ψ_{rms} [Wb]	E_{rms} [V]
a. No-load	-0.2361	0.2914	91.5568
b. Nominal current with $\delta = 0^\circ$	-0.1743	0.2993	94.0404
c. Nominal current with $\delta = 90^\circ$	14.4107	0.2918	91.6634
d. Nominal current with $\delta = 180^\circ$	-0.2361	0.2914	91.5568

Table 3: Torque, flux linkage and induced voltage for different load angles

B.4 Air-Gap Magnetic Flux Density

This section presents the air-gap magnetic flux density B plots and the corresponding harmonic analysis (FFT) for the operating conditions examined in Question B.2.

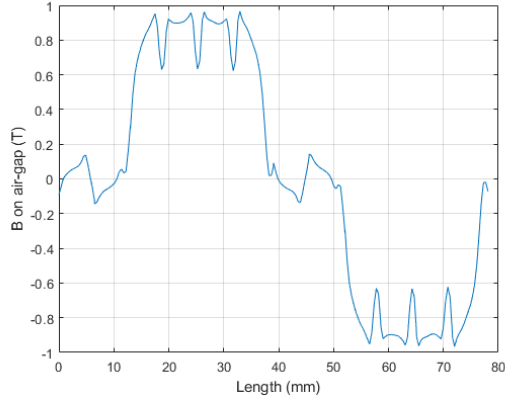


Figure 14: Magnetic Flux Density for 0 load

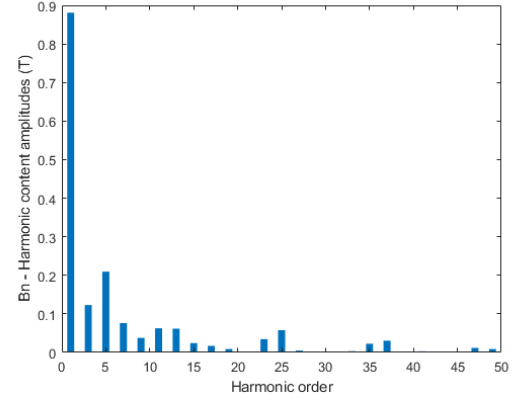


Figure 15: FFT of Magnetic Flux Density for 0 load

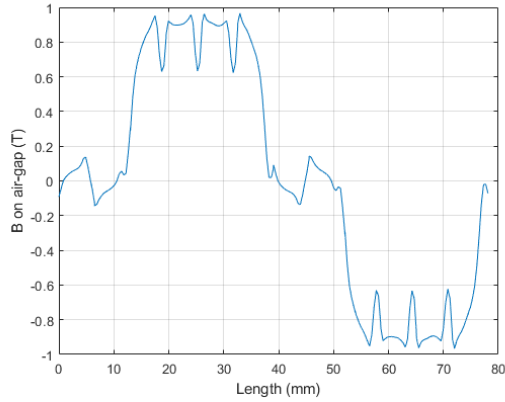


Figure 16: Magnetic Flux Density for $\delta = 0^\circ$

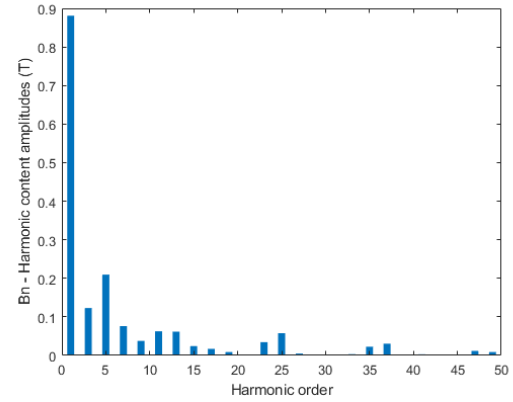


Figure 17: FFT of Magnetic Flux Density for $\delta = 0^\circ$

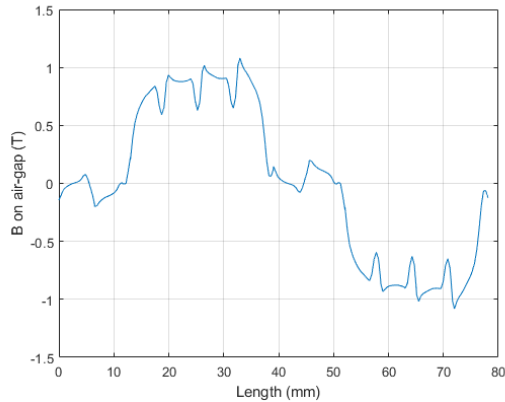


Figure 18: Magnetic Flux Density for $\delta = 90^\circ$

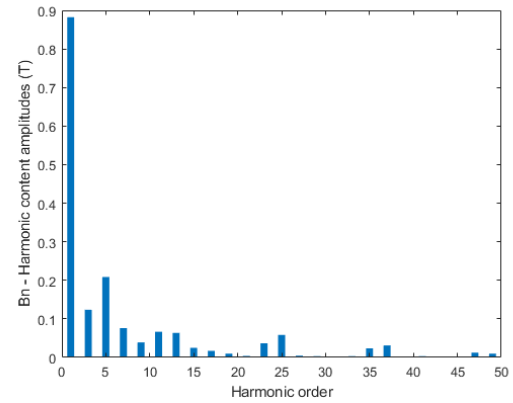


Figure 19: FFT of Magnetic Flux Density for $\delta = 90^\circ$

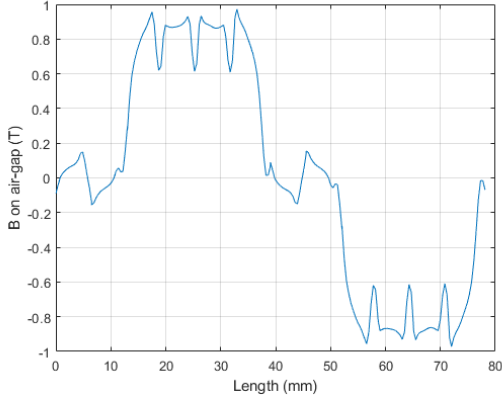


Figure 20: Magnetic Flux Density for $\delta = 180^\circ$

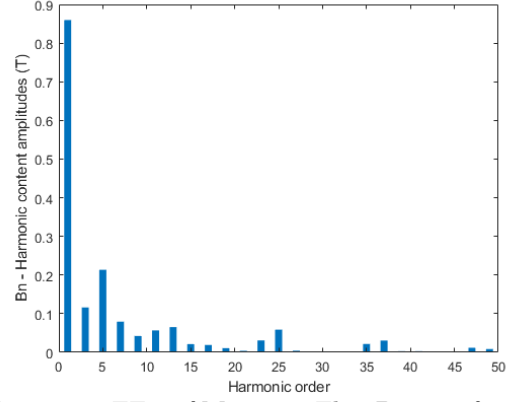


Figure 21: FFT of Magnetic Flux Density for $\delta = 180^\circ$

B.5 Calculation of d- and q-axis Inductances L_d and L_q

To determine the inductances L_d and L_q , the machine quantities were transformed from the natural three-phase (abc) coordinate system to the rotating (dq) reference frame.

First, the Clarke transformation was used to move to the stationary two-axis $\alpha\beta$ reference frame:

$$\begin{bmatrix} i_\alpha \\ i_\beta \end{bmatrix} = \frac{2}{3} \begin{bmatrix} 1 & -1/2 & -1/2 \\ 0 & \sqrt{3}/2 & -\sqrt{3}/2 \end{bmatrix} \begin{bmatrix} i_a \\ i_b \\ i_c \end{bmatrix} \quad (26)$$

Next, the $\alpha\beta$ system is rotated by the electrical angle θ_e to align with the rotor magnet axis (the d -axis):

$$\begin{bmatrix} i_d \\ i_q \end{bmatrix} = \begin{bmatrix} \cos(\theta_e) & \sin(\theta_e) \\ -\sin(\theta_e) & \cos(\theta_e) \end{bmatrix} \begin{bmatrix} i_\alpha \\ i_\beta \end{bmatrix} \quad (27)$$

After the appropriate transformations, the values of the currents i_d, i_q , the corresponding flux linkages Ψ_d, Ψ_q , and the permanent-magnet flux Ψ_m allow the machine parameters to be calculated:

d-axis inductance:

$$L_d = \frac{\Psi_d - \Psi_m}{i_d} = 1.6508mH \quad (28)$$

q-axis inductance:

$$L_q = \frac{\Psi_q}{i_q} = 3.7218mH \quad (29)$$

It should be noted that:

- The d- and q-axis inductances are not defined under no-load conditions because $i_d, i_q = 0$.
- The measurement of Ψ_m was performed under no-load conditions, since in this case $\Psi_m = \Psi_d$ when phase A is aligned with the d-axis.
- The measurements for L_q were performed at an internal torque angle of $\delta = 90^\circ$, since i_q reaches its maximum value and $i_d = 0$.

- Similarly, the measurements for L_d were performed at an internal torque angle of $\delta = 0^\circ$, since i_d reaches its maximum value and $i_q = 0$.

It is observed that, although the designed machine is a surface-mounted permanent-magnet synchronous machine (SPMSM) and is theoretically classified as a salient-pole machine, it exhibits relatively strong saliency ($L_d \neq L_q$).

B.6 Magnetic Field Distribution for Different Current Densities

The following plots show the magnetic flux density B for different current-density values J , while keeping the internal torque angle fixed at $\delta = 90^\circ$. The following operating conditions are considered:

- In Figure 22, half the nominal current density: $J = J_N/2 = 2A/mm^2$.
- In Figure 23, nominal current density: $J_N = 4A/mm^2$.
- In Figure 24, 1.5 times the nominal current density: $J = 1.5 \cdot J_N = 6A/mm^2$.

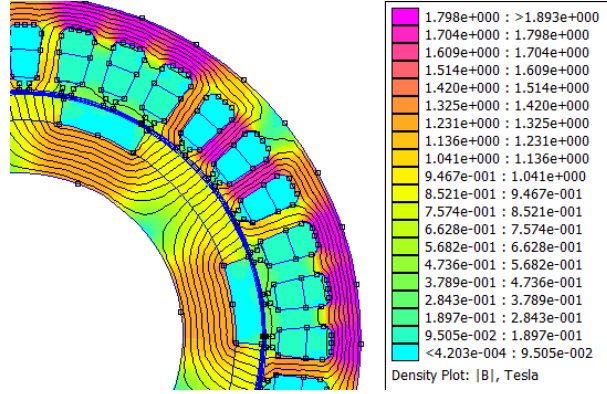


Figure 22: Magnetic Flux Density for $J = 2A/mm^2$

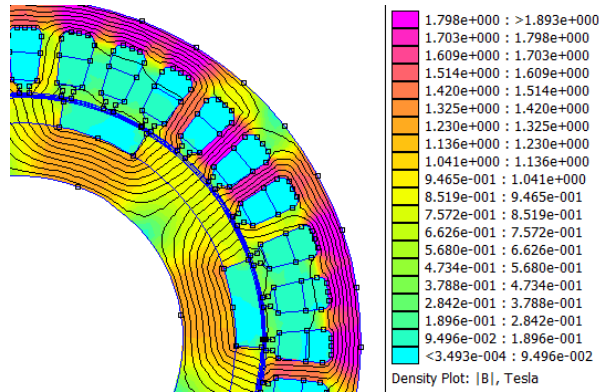


Figure 23: Magnetic Flux Density for $J = 4A/mm^2$

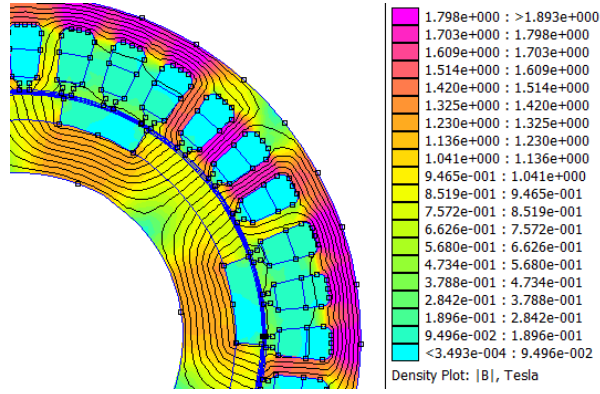


Figure 24: Magnetic Flux Density for $J = 8 \text{ A/mm}^2$

B.7 Operating Points of the Ferromagnetic Material on the $B - H$ Curve

To visualize the two operating points of the ferromagnetic material for the different load conditions of Question B.6, the midpoint of the tooth and the midpoint of the stator yoke were selected, as shown in Figure 25.

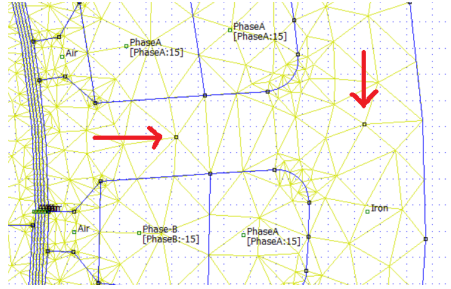


Figure 25: Tooth and Yoke points selected for the B-H curve

The magnetic flux density B and magnetic field strength H were measured at these points and plotted on the $B - H$ curve. Figures 26 and 27 show the two curves for two different torque angles as representative cases.

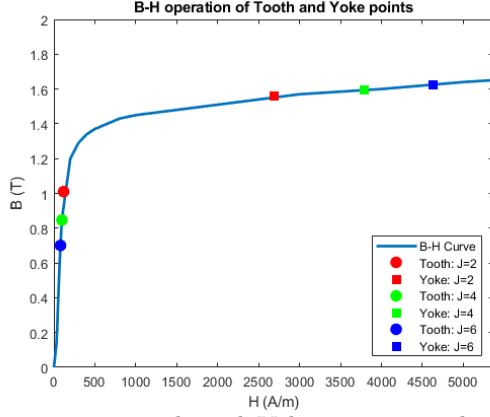


Figure 26: Tooth and Yoke points on the B-H curve

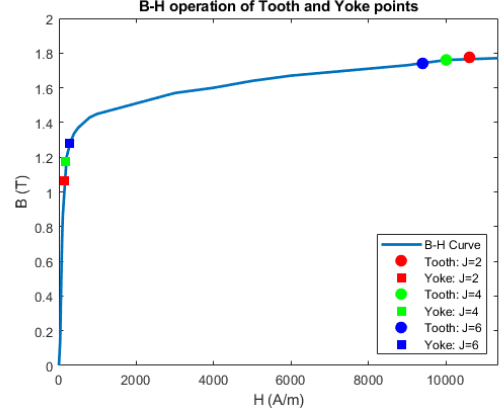


Figure 27: Tooth and Yoke points on the B-H curve for a different δ

From the operating points plotted on the magnetization curve $B - H$ for current densities $J = 2, 4, 6 \text{ A/mm}^2$ in Figure 26, it follows that the stator yoke operates deep in saturation, with flux-density values exceeding 1.6 T. In contrast, the selected tooth point remains in the linear region of the curve with very low flux density ($B < 0.2 \text{ T}$), indicating that under this particular operating condition of the machine (rotor position and stator current angle), the magnetic flux closes mainly through the yoke. It is also observed that increasing the current density shifts the yoke operating points toward higher magnetic-field strength H and consequently higher flux-density B , while the tooth operating points exhibit the opposite behavior.

However, not all teeth and all points in the yoke behave in the same way. For this reason, Figure 27 presents the same representation for a different torque angle. In this case, the selected tooth point operates in the saturation region, whereas the yoke operates in the linear region.

C Supplementary Study with Synchronous Rotation

C.1 Sweep of Internal Current Angles and Current Magnitudes

The analysis that follows was performed by sweeping the internal current angle and the magnitude of the machine supply current. Specifically, the problem was solved for 10 different values of the current angle δ and 10 different current magnitudes I . For each combination of these parameters, the rotor was synchronously rotated with an electrical-angle step of 3° to obtain the electromagnetic torque in the machine air gap, the winding flux linkages, as well as the magnetic vector potential at selected rotor and stator points.

For the system analysis, the dq reference frame was used through the Clarke and Park transformations, since the study concerns synchronous rotor rotation. The choice of this reference frame significantly simplifies the analysis, as it transforms the three-phase quantities into constant components in the reference frame rotating at synchronous speed.

The internal current angle δ was defined in the dq reference frame as the angle between the current vector and the q -axis, as shown below:

$$\delta = \arctan\left(-\frac{i_d}{i_q}\right). \quad (30)$$

Thus, the current components are given by:

$$i_q = I \cos(\delta), \quad i_d = -I \sin(\delta). \quad (31)$$

It should be noted that this current angle corresponds to the internal torque angle used in the previous questions, with a difference of 90° . The sweep range of the internal current angle was limited to $0^\circ \leq \delta \leq 90^\circ$, while the current magnitude was varied from zero to the maximum permissible value $I_{\max} = 4.7931$ A (without taking the fill factor into account).

Finally, to increase the data density and obtain smoother plots of the quantities, an interpolation procedure was applied, resulting in a tenfold increase in the number of available data points.

C.2 Torque, Flux Linkage, and Induced Voltage Maps

In Figure 28, the electromagnetic torque map is presented as a function of current magnitude and angle δ .

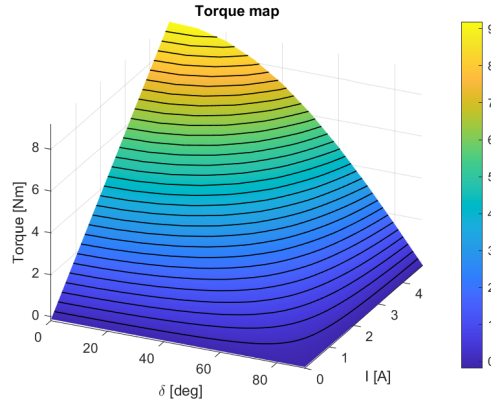


Figure 28: Torque map as a function of current magnitude and internal current angle δ

Figure 29 shows the corresponding phase-A flux-linkage map, which illustrates the variation of flux linkage with current magnitude and current angle.

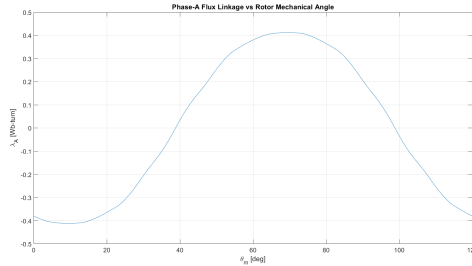


Figure 29: Phase-A flux linkage map as a function of current magnitude and internal current angle δ

Figures 30a, 30b, and 30c show the phase-A induced-voltage maps for rotational speeds of 500, 1000, and 2000 rpm, respectively.

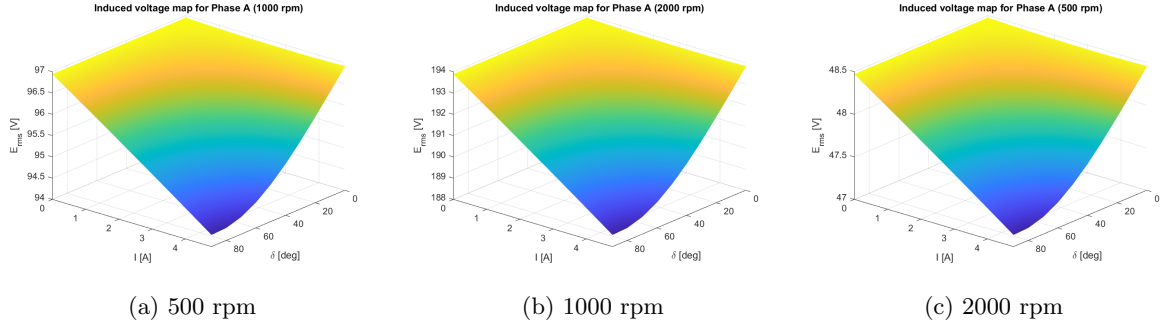


Figure 30: Phase-A induced voltage maps for 500, 1000 and 2000 rpm

The maps show that electromagnetic torque increases with increasing current magnitude, while it decreases as the internal current angle δ increases. In contrast, flux linkage increases for smaller current magnitudes or for larger values of the angle δ , which is related to the redistribution of the current components along the d and q axes.

The observed behavior of the maps can be interpreted using the fundamental equations of the permanent-magnet synchronous machine in the dq reference frame. For an SPMSM, the electromagnetic torque is given by:

$$T_e = \frac{3}{2} p \Psi_m i_q \quad (32)$$

where Ψ_m is the permanent-magnet flux and i_q is the current component along the q -axis.

It follows that torque increases with the current magnitude I , while it decreases for increasing values of the angle δ , as the q -axis current component decreases.

The flux linkages along the d - and q -axes are given by, respectively:

$$\Psi_d = \Psi_m + L_d i_d \quad \Psi_q = L_q i_q \quad (33)$$

As the angle δ increases, i_d becomes more negative, and the total flux linkage increases, which explains the increase in Ψ for larger values of δ or for smaller current magnitudes.

C.3 Determination of the MTPA Curve

The MTPA (Maximum Torque Per Ampere) strategy aims to identify the operating points at which the maximum electromagnetic torque is achieved for a given current magnitude.

Figure 31 shows the electromagnetic torque map, together with the MTPA curve without a voltage constraint, obtained by selecting, for each current magnitude, the angle δ that maximizes torque.

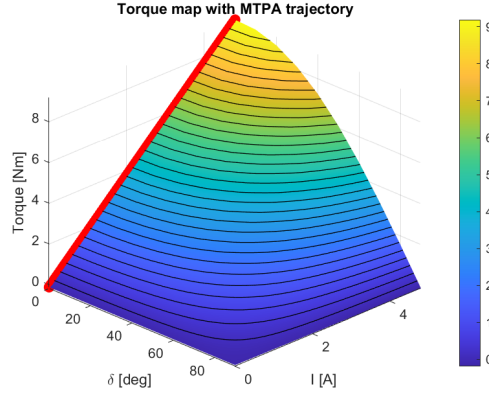


Figure 31: Torque map with MTPA trajectory

Figure 32 shows the MTPA angle as a function of current magnitude. At the same time, Figure 33 shows the maximum torque achieved for each current magnitude along the MTPA curve. It is observed that the optimum angle for this motor is 0° , and the maximum torque varies linearly with current magnitude, which is expected since an SPMSM is being considered.

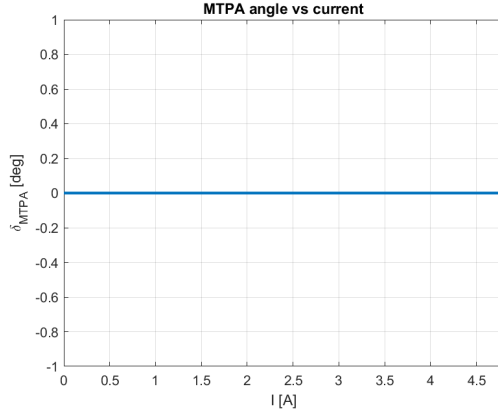


Figure 32: MTPA angle as a function of current magnitude

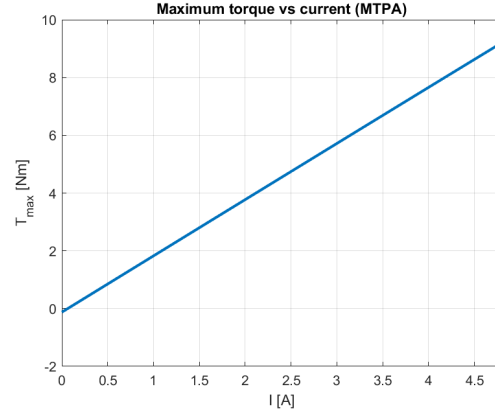


Figure 33: Maximum torque versus current along the MTPA trajectory

C.4 Imposition of a Voltage Constraint and Its Effect on Operating Points

In this question, a constraint is imposed on the induced voltage of the machine and its effect on the feasible operating points and on the optimal MTPA current-angle values is investigated. The analysis is performed at two rotational speeds: the nominal speed of 1000 rpm and twice this speed, 2000 rpm.

For a speed of 1000 rpm, the induced-voltage limit was set to $E_{\text{rms}} = 96.5$ V, while for 2000 rpm, the corresponding limit was set to $E_{\text{rms}} = 195$ V. These values were selected to represent the voltage constraint at different rotational-speed levels. It should be noted that two different voltage limits are selected because, with the given geometry and parameters, using the same limit while doubling the speed would result in zero torque in all cases. for different rotational-speed levels. It should be noted that two different voltage limits are selected because, with the given geometry and parameters, using the same limit while doubling the speed would result in zero torque in all cases.

Figures 34 and 35 show the torque map with the voltage-limited MTPA trajectory for the two rotational speeds.

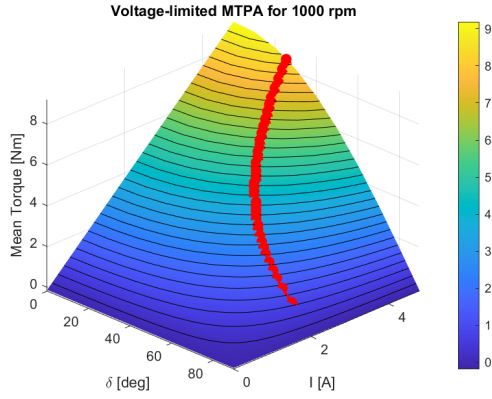


Figure 34: Voltage-limited MTPA trajectory for 1000 rpm

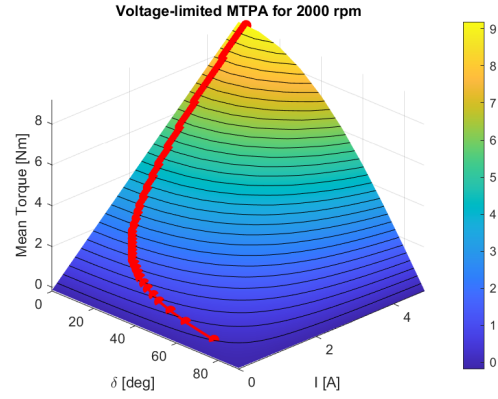


Figure 35: Voltage-limited MTPA trajectory for 2000 rpm

Figures 36 and 37 show the feasible torque region under the voltage constraint.

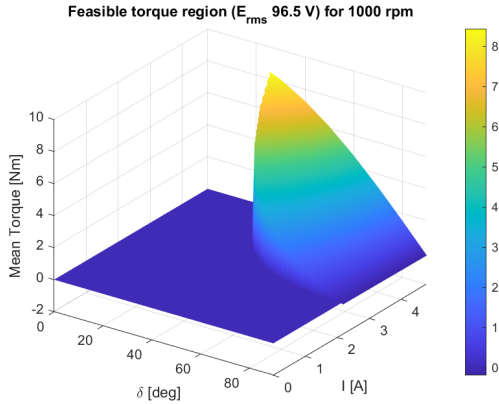


Figure 36: Feasible torque region for 1000 rpm

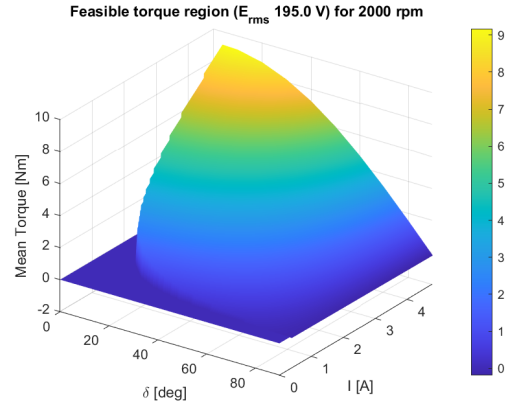


Figure 37: Feasible torque region for 2000 rpm

Figures 38 and 39 show the voltage limit on the MTPA curve in the current-magnitude- δ plane for the two rotational speeds. Similarly, Figures 41 and 40 show the maximum torque achieved for each current magnitude along the MTPA curve with the corresponding voltage constraint. It is observed that the more restrictive the constraint, i.e., in the case of 1000 rpm, the more the curve deviates from the linear relationship.

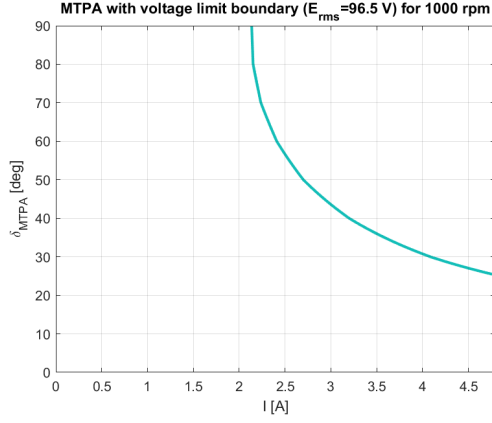


Figure 38: MTPA - Voltage limit boundary for 1000 rpm

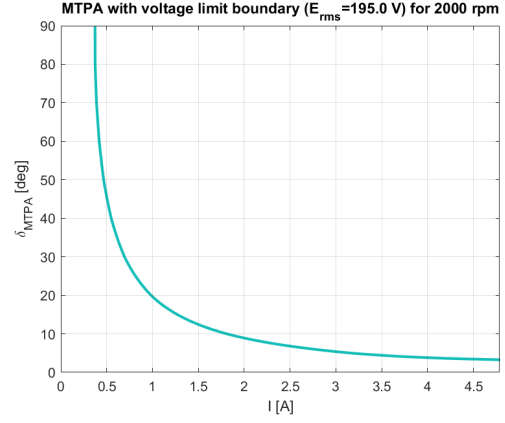


Figure 39: MTPA - Voltage limit boundary for 2000 rpm

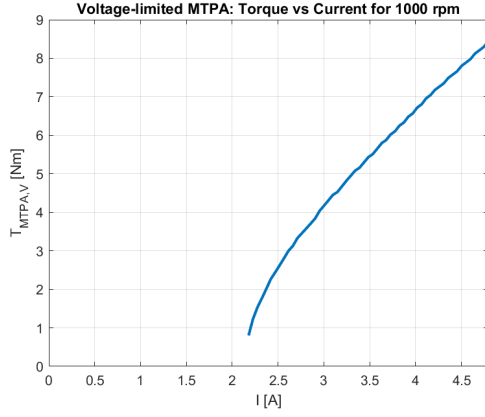


Figure 40: Voltage-limited MTPA torque versus current for 1000 rpm

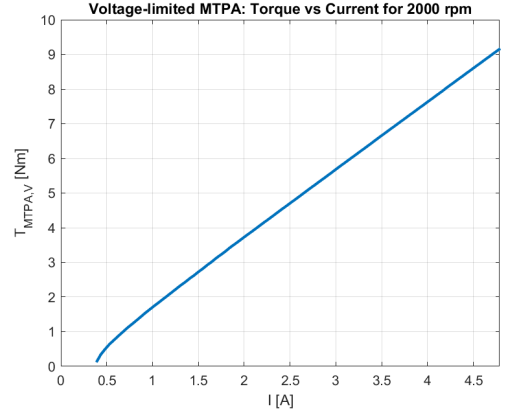


Figure 41: Voltage-limited MTPA torque versus current for 2000 rpm

C.5 Magnetic-Field Variation and Loci of Magnetic Flux Density

In this question, the variation of the magnetic field at selected points in the machine core, both in the stator and rotor, during synchronous rotation is studied. A nominal current of $I = I_n = 4.7931$ and a current angle of $\delta = 0^\circ$ are used. The points selected for visualization in the stator and rotor cores are shown in Figure 42.

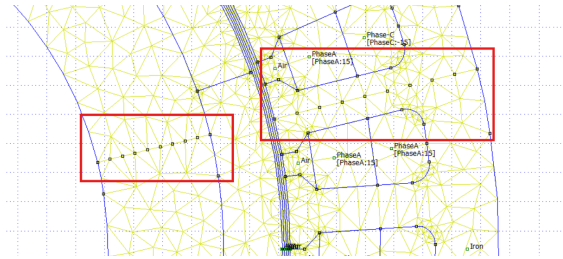


Figure 42: Selected stator and rotor points

Figure 43 shows the variation in the magnitude of the magnetic flux density $|\mathbf{B}|$ at selected points in the stator core as a function of electrical angle. Note that “Point 1” corresponds to the point with the smallest radius, while “Point 10” corresponds to the point with the largest radius.

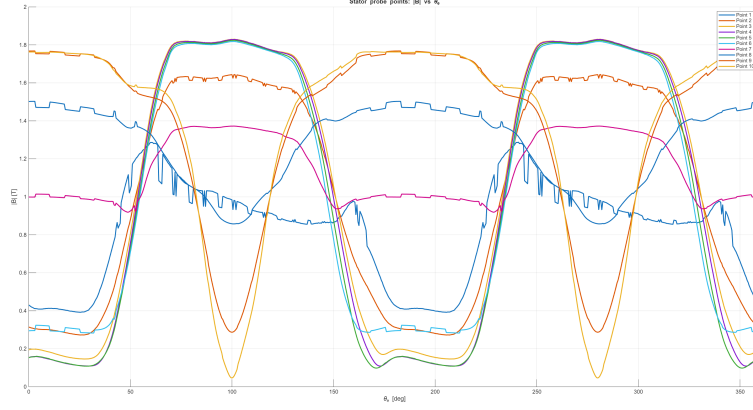


Figure 43: Magnitude of magnetic flux density at stator probe points

Figure 44 shows the loci of the magnetic flux-density vector in the (B_x, B_y) plane for the same stator points.

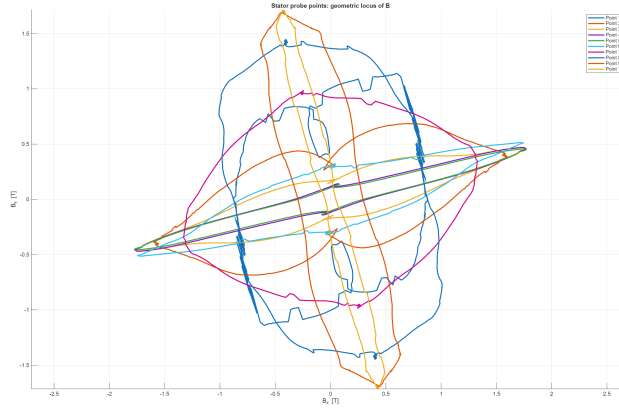


Figure 44: B_x – B_y plane for the stator points

Similarly, Figure 45 shows the variation of the magnetic flux-density magnitude at selected points in the rotor core as a function of electrical angle.

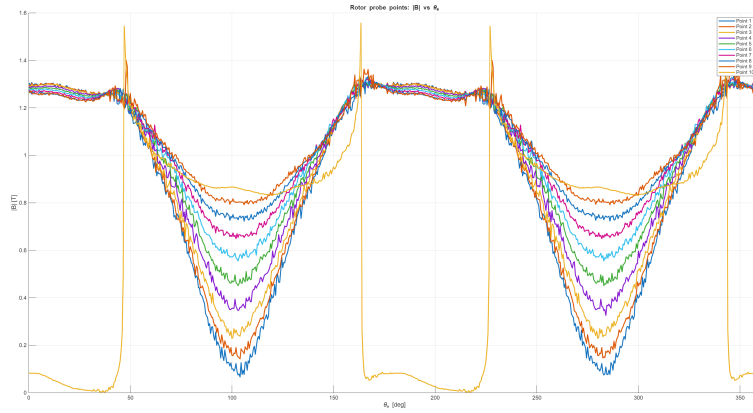


Figure 45: Magnitude of magnetic flux density at rotor probe points

Finally, Figure 46 shows the loci of the magnetic flux-density vector in the (B_x, B_y) plane for the rotor points.

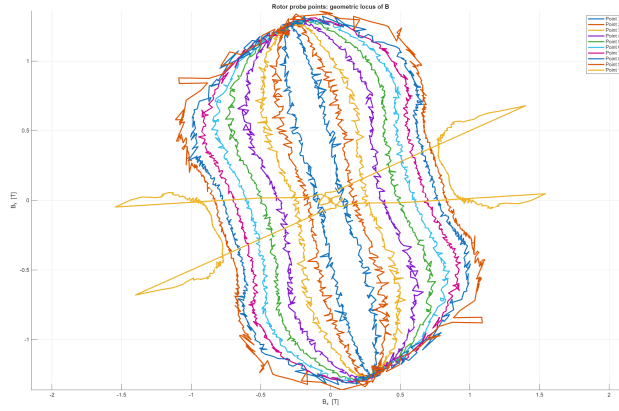


Figure 46: B_x - B_y plane for the rotor points

C.6 Torque, Flux-Linkage, and Induced-Voltage Waveforms

In this question, the waveforms of electromagnetic torque, flux linkage, and induced voltage in the motor windings are presented as a function of electrical rotation angle, for two characteristic operating conditions: nominal operation at $I = I_n = 4.9731$ and overload at $I = 2 \cdot I_n = 9.9462$ at the nominal speed $N_n = 1000$ rpm.

Figures 47 and 48 show the flux-linkage waveforms of the three phases for nominal operating conditions and overload conditions, respectively.

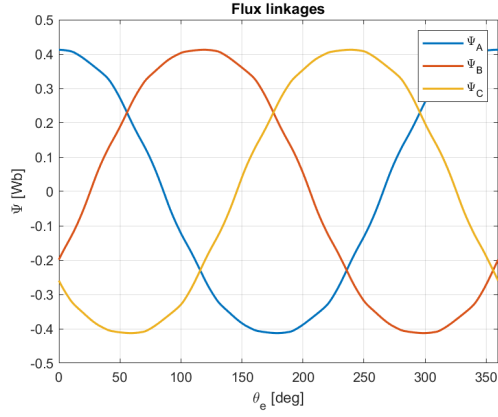


Figure 47: Flux linkages for nominal operating conditions

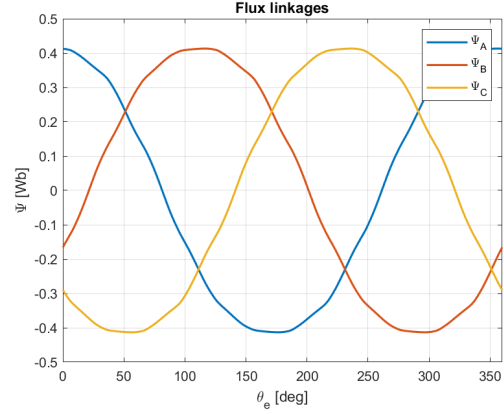


Figure 48: Flux linkages for overload conditions

Figures 49 and 50 show the induced-voltage waveforms (Back-EMF) of the induced voltage (Back-EMF) for nominal operation and overload conditions, respectively.

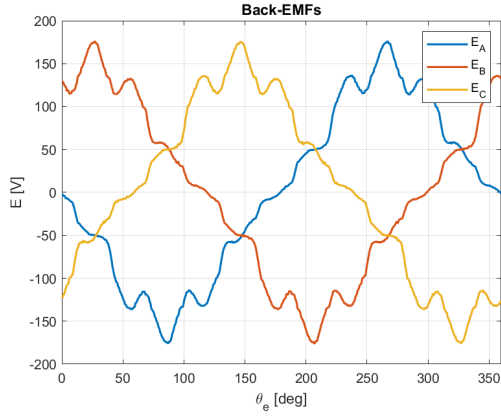


Figure 49: Back-EMF waveforms for nominal operating conditions

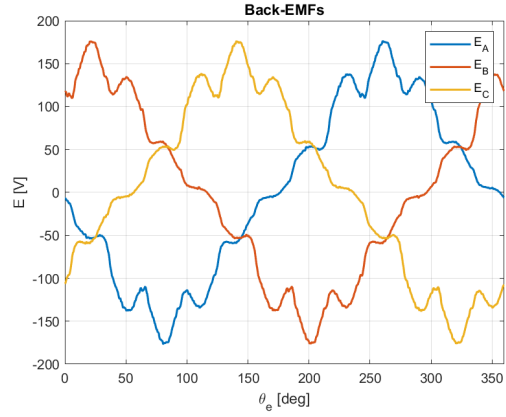


Figure 50: Back-EMF waveforms for overload conditions

Figures 51 and 52 show the electromagnetic torque-ripple waveform as a function of electrical angle for nominal and overload conditions, respectively.

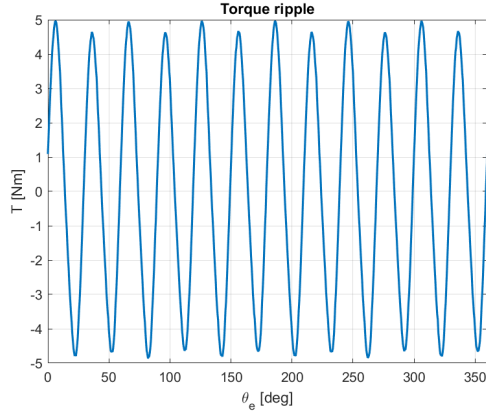


Figure 51: Torque ripple for nominal operating conditions

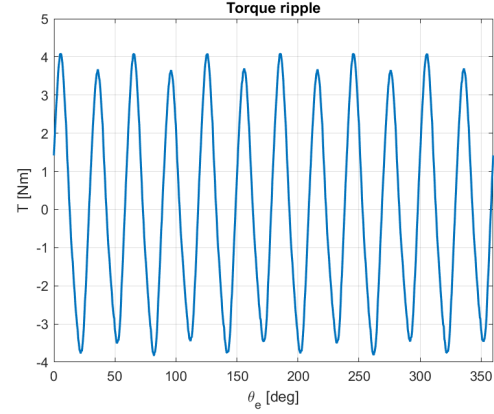


Figure 52: Torque ripple for overload conditions

C.7 Torque Ripple and Harmonic Content

In this question, the electromagnetic torque ripple is calculated, as well as the harmonic content of the flux linkage and induced voltage for one motor phase, for nominal and overload operation.

From Figures 51 and 52, the torque ripple under nominal and overload conditions, respectively, is:

$$T_{\text{ripple},n} = 9.824 \text{ Nm} \quad T_{\text{ripple},ov} = 7.915 \text{ Nm} \quad (34)$$

Figures 53 and 54 show, for comparison, the harmonic content of the phase-A flux linkage for nominal operation and overload operation.

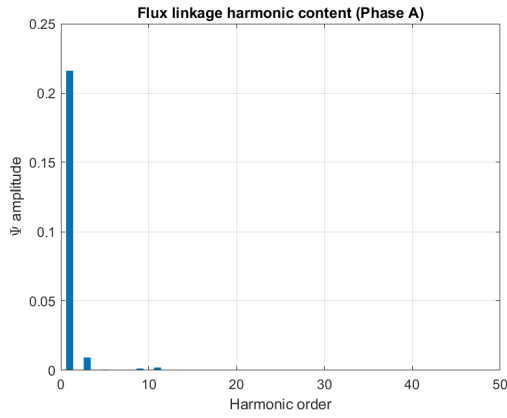


Figure 53: Flux linkage harmonic content for nominal operating conditions

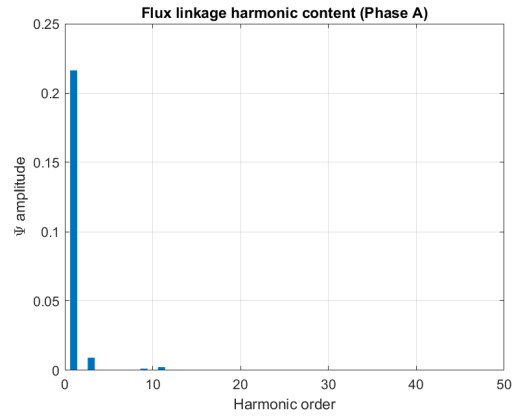


Figure 54: Flux linkage harmonic content for overload conditions

Similarly, Figures 55 and 56 show the harmonic content of the phase-A induced voltage for the two operating conditions.

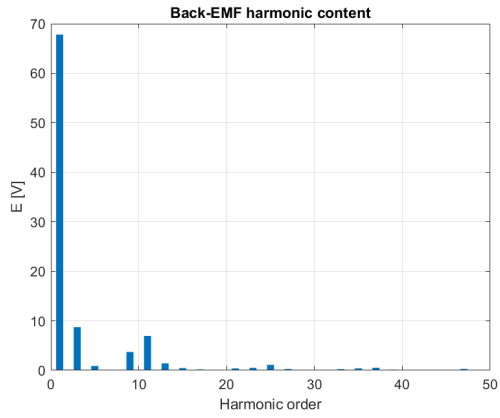


Figure 55: Back-EMF harmonic content for nominal operating conditions

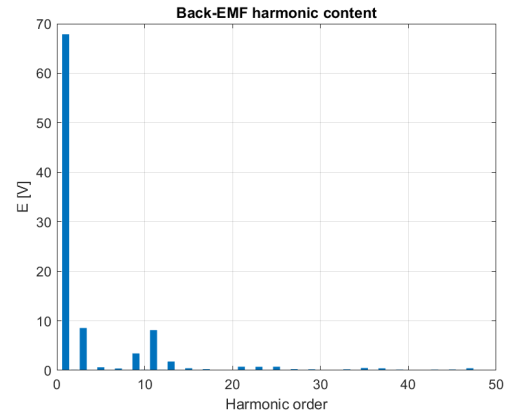


Figure 56: Back-EMF harmonic content for overload conditions